

13 Vector Field, Divergence and Curl

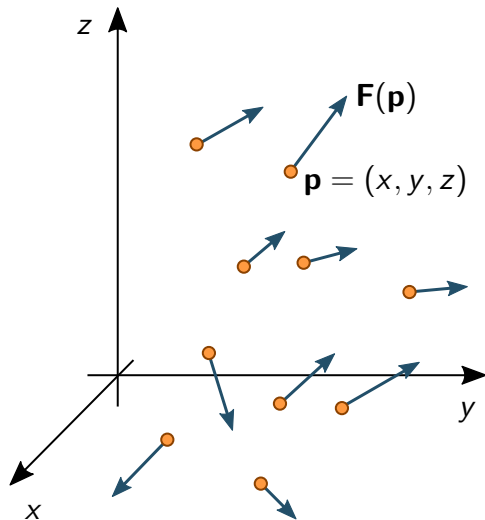
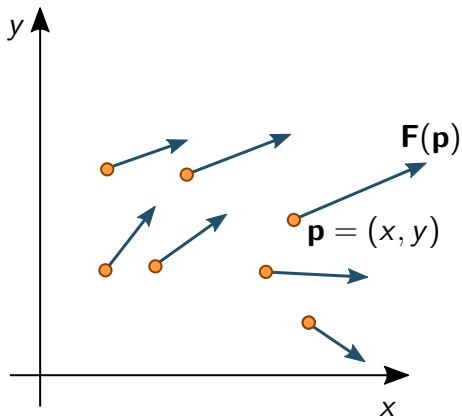
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Vector Field

A mapping $\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ can be called a vector field. We visualise vector fields as vectors attached to points in space. These vectors correspond to the values of $\mathbf{F}$.



Notation for vector fields

A vector field $\mathbf{F}$ defined on $\mathbb{R}^3$ can be written as

$$\begin{aligned}\mathbf{F}(\mathbf{p}) &= \mathbf{F}(x, y, z) \\ &= (F_1(x, y, z), F_2(x, y, z), F_3(x, y, z)) \\ &= F_1(x, y, z)\mathbf{i} + F_2(x, y, z)\mathbf{j} + F_3(x, y, z)\mathbf{k},\end{aligned}$$

where $\mathbf{p} = (x, y, z)$.

Likewise, a vector field $\mathbf{F}$ on $\mathbb{R}^2$ is

$$\begin{aligned}\mathbf{F}(\mathbf{p}) &= \mathbf{F}(x, y) \\ &= (F_1(x, y), F_2(x, y)) \\ &= F_1(x, y)\mathbf{i} + F_2(x, y)\mathbf{j},\end{aligned}$$

where $\mathbf{p} = (x, y)$.

Example 13.1

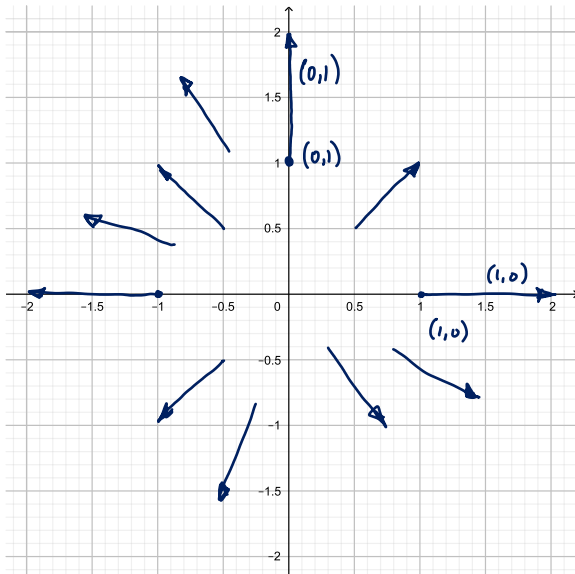
$$\operatorname{div} \underline{F} = \nabla \cdot \underline{F} = \frac{\partial}{\partial x} \cdot x + \frac{\partial}{\partial y} \cdot y = 2 > 0$$

$$\operatorname{curl} \underline{F} = 0$$

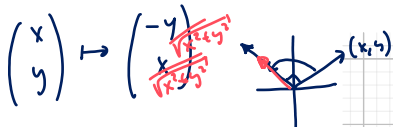
Sketch some vectors in the plane associated with the vector field

$$\mathbf{F}(x, y) = (x, y) = x\mathbf{i} + y\mathbf{j}.$$

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x \\ y \end{pmatrix}$$



Example 13.2

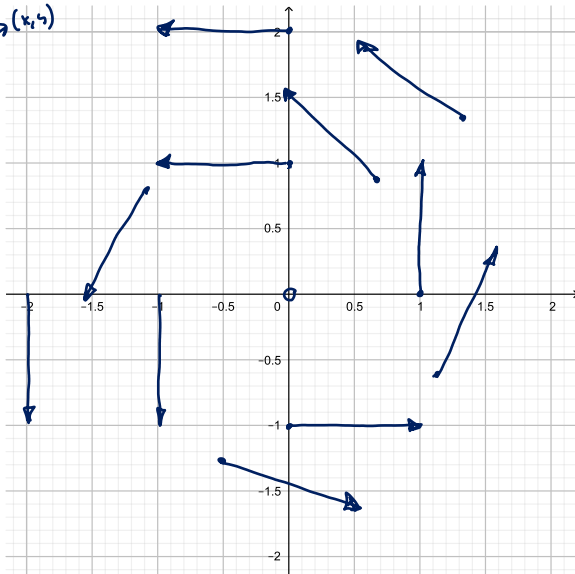


Sketch some vectors in the plane associated with the vector field

$$\begin{aligned}\mathbf{F}(x, y) &= \left(\frac{-y}{\sqrt{x^2 + y^2}}, \frac{x}{\sqrt{x^2 + y^2}} \right) \\ &= \frac{-y}{\sqrt{x^2 + y^2}} \mathbf{i} + \frac{x}{\sqrt{x^2 + y^2}} \mathbf{j},\end{aligned}$$

where $(x, y) \neq (0, 0)$. $\operatorname{div} \mathbf{F} = 0$

$$\operatorname{curl} \mathbf{F} = \frac{1}{\sqrt{x^2 + y^2}}$$



Del operator

The ∇ (called 'del' or 'nabla') operator:

$$\nabla := \left(\frac{\partial}{\partial x} \mathbf{i} + \frac{\partial}{\partial y} \mathbf{j} + \frac{\partial}{\partial z} \mathbf{k} \right)$$

Recall that ∇ defines the gradient of a function $\phi : \mathbb{R}^3 \rightarrow \mathbb{R}$,

$$\nabla \phi = \left(\frac{\partial}{\partial x} \mathbf{i} + \frac{\partial}{\partial y} \mathbf{j} + \frac{\partial}{\partial z} \mathbf{k} \right) \phi = \frac{\partial \phi}{\partial x} \mathbf{i} + \frac{\partial \phi}{\partial y} \mathbf{j} + \frac{\partial \phi}{\partial z} \mathbf{k}.$$

This operator is also used to define the **divergence** and **curl** of a vector field.

Definition 13.3

The divergence of a differentiable vector field $\mathbf{F} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$ is

$$\operatorname{div} \mathbf{F} = \underline{\nabla \cdot \mathbf{F}} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}.$$

$$\nabla \cdot \mathbf{F} = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \cdot \begin{pmatrix} F_1 \\ F_2 \\ F_3 \end{pmatrix} = \boxed{\frac{\partial}{\partial x} F_1 + \frac{\partial}{\partial y} F_2 + \frac{\partial}{\partial z} F_3}$$



Definition 13.4

The curl of a differentiable vector field $\mathbf{F} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$ is

$$\operatorname{curl} \mathbf{F} = \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix} = \underbrace{\left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right) \mathbf{i} + \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \right) \mathbf{j} + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \mathbf{k}}.$$

For $\mathbf{F} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ we let $\operatorname{curl} \mathbf{F} = \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right)$.

Example 13.5

Let $\mathbf{F}(x, y, z) = y^3\mathbf{i} + 2yz\mathbf{j} - xz^2\mathbf{k}$. $= \begin{pmatrix} y^3 \\ 2yz \\ -xz^2 \end{pmatrix}$

(a) Evaluate $\text{div } \mathbf{F}$ and $\text{curl } \mathbf{F}$.

(b) Evaluate $\text{div } \mathbf{F}$ and $\text{curl } \mathbf{F}$ at $(2, 1, -1)$.

$$(a) \text{div } \underline{\mathbf{F}} = \nabla \cdot \underline{\mathbf{F}} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} = 0 + 2z - 2xz = \underline{2z(1-x)}$$

$$\text{curl } \underline{\mathbf{F}} = \nabla \times \underline{\mathbf{F}} = \begin{pmatrix} \frac{\partial F_2}{\partial x} - \frac{\partial F_3}{\partial y} \\ \frac{\partial F_3}{\partial z} - \frac{\partial F_1}{\partial x} \\ \frac{\partial F_1}{\partial y} - \frac{\partial F_2}{\partial z} \end{pmatrix} = \begin{pmatrix} 0 - 2y \\ 0 + z^2 \\ 0 - 3y^2 \end{pmatrix} = \begin{pmatrix} -2y \\ +z^2 \\ -3y^2 \end{pmatrix}$$

$$b) \text{div } \underline{\mathbf{F}}(2, 1, -1) = 2 \cdot (-1) \cdot (1 - 2) = 2$$

$$\text{curl } \underline{\mathbf{F}}(2, 1, -1) = (-2, 1, -3)$$

Curl in two dimensions

Example 13.6

For $\mathbf{F}(x, y) = y \cos x \mathbf{i} + \sin y \mathbf{j}$ calculate $\operatorname{div} \mathbf{F}$ and $\operatorname{curl} \mathbf{F}$.

$$\operatorname{curl} \mathbf{F} = \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} = 0 - \cos x = -\cos x$$

$$\operatorname{div} \mathbf{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} = -y \sin x + \cos y$$

Some useful relations

$$\psi: \mathbb{R}^3 \rightarrow \mathbb{R}, \quad \varphi: \mathbb{R}^2 \rightarrow \mathbb{R}$$

$$\underline{F}: \mathbb{R}^3 \rightarrow \mathbb{R}^3, \quad \underline{F}: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$\operatorname{div} \underline{F} = \sum_{i=1}^n \frac{\partial F_i}{\partial x_i} \quad - \text{ans dimension}$$

$$\operatorname{curl} \underline{F} = \nabla \times \underline{F}$$

$$\operatorname{curl}(\nabla \psi) = 0$$

$$\boxed{\nabla \times (\nabla \phi) = 0}$$

$$\psi = \begin{pmatrix} \frac{\partial \psi}{\partial x} \\ \frac{\partial \psi}{\partial y} \\ \frac{\partial \psi}{\partial z} \end{pmatrix}$$

Let ϕ be a (scalar) function and let $\mathbf{F}$ be a vector field.

- If ϕ has continuous second partial derivatives, then

- If components of $\mathbf{F}$ have continuous second partial derivatives, then

$$\underline{\nabla \cdot (\nabla \times \mathbf{F})} = \underline{\nabla \cdot \operatorname{curl} \mathbf{F}} = \underline{0}.$$

$$F_1 = \frac{\partial \psi}{\partial x}, \quad F_2 = \frac{\partial \psi}{\partial y}, \quad F_3 = \frac{\partial \psi}{\partial z}$$

$$\frac{\partial F_2}{\partial y} - \frac{\partial F_1}{\partial z} = 0$$

$$= \frac{\partial}{\partial y} \frac{\partial \psi}{\partial z} - \frac{\partial}{\partial z} \frac{\partial \psi}{\partial y}$$

Product rule for divergence

$$\underline{\nabla \cdot (\phi \mathbf{F})} = (\nabla \phi) \cdot \mathbf{F} + \phi \cdot (\nabla \cdot \mathbf{F})$$

$$\operatorname{div}(\phi \mathbf{F}) = (\nabla \phi) \cdot \mathbf{F} + \phi \cdot (\operatorname{div} \mathbf{F})$$

Product rule for curl

$$\nabla \times (\phi \mathbf{F}) = (\nabla \phi) \times \mathbf{F} + \phi \cdot (\nabla \times \mathbf{F})$$

$$\operatorname{curl}(\phi \mathbf{F}) = (\nabla \phi) \times \mathbf{F} + \phi \cdot (\operatorname{curl} \mathbf{F})$$

$$\nabla \cdot (\psi \underline{F}) = \frac{\partial}{\partial x} (\psi F_1) + \frac{\partial}{\partial y} (\psi F_2) + \frac{\partial}{\partial z} (\psi F_3)$$

$$= \frac{\partial}{\partial x} (\psi \cdot F_1) + \frac{\partial}{\partial y} (\psi F_2) + \frac{\partial}{\partial z} (\psi F_3) = \left(\frac{\partial}{\partial x} \psi \right) F_1 + \psi \frac{\partial}{\partial x} F_1 + \frac{\partial \psi}{\partial y} F_2 + \psi \frac{\partial}{\partial y} F_2 + \frac{\partial \psi}{\partial z} F_3 + \psi \frac{\partial}{\partial z} F_3$$

$$= \nabla \psi \cdot \underline{F} + \psi \cdot (\nabla \cdot \underline{F})$$

Example 13.7

Let $\phi = y - x + z$ and $\mathbf{F} = \underline{z}\mathbf{i} + \underline{y}\mathbf{j} + \underline{x}\mathbf{k}$. Verify the product rule for divergence for this example.

$$\underbrace{\nabla \cdot (\phi \mathbf{F})}_{LHS} = \underbrace{\nabla \phi \cdot \mathbf{F} + \phi \cdot (\nabla \cdot \mathbf{F})}_{RHS}$$

$$G = \phi \mathbf{F}$$

$$LHS = \nabla \cdot \left[(y-x+z) \cdot \begin{pmatrix} z \\ y \\ x \end{pmatrix} \right] = \nabla \cdot \begin{bmatrix} \frac{\partial}{\partial x} [(y-x+z)z] \\ \frac{\partial}{\partial y} [(y-x+z)y] \\ \frac{\partial}{\partial z} [(y-x+z)x] \end{bmatrix} = \frac{\partial \phi}{\partial x} z + \frac{\partial \phi}{\partial y} y + \frac{\partial \phi}{\partial z} x = \underline{2y}$$

$$RHS = \nabla \phi \cdot \mathbf{F} + \phi \cdot (\nabla \cdot \mathbf{F}) = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} z \\ y \\ x \end{pmatrix} + (y-x+z) \cdot (1)$$

$$= -z + y + x + y - x + z = 2y$$

$$LHS = RHS \quad \square$$

Conservative and irrotational vector fields; scalar potential

Definition 13.8 (Conservative vector field)

A vector field $\mathbf{F}$ is called conservative (or potential or gradient vector field) if there exists a scalar function ϕ such that $\nabla\phi = \mathbf{F}$.

The function ϕ is called the scalar potential of $\mathbf{F}$.

Example 13.9

The gravitational field of a mass M at the origin acting on a test mass m at position $\mathbf{r}$,

$$\mathbf{F} = -G \frac{mM}{\|\mathbf{r}\|^3} \mathbf{r} = -G \frac{mM}{(x^2+y^2+z^2)^{3/2}} \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad \mathbf{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$
$$\phi(\mathbf{r}) = \frac{c}{\sqrt{x^2+y^2+z^2}}$$

is conservative.

$$\phi(\mathbf{r}) = \frac{GmM}{\|\mathbf{r}\|}$$

$$c = GmM$$

$$\nabla\phi(\mathbf{r}) = \nabla\phi(x,y,z) = \begin{bmatrix} c \cdot \frac{1}{2} \cdot \frac{2x}{(\sqrt{x^2+y^2+z^2})^3} & -\frac{cy}{\sqrt{x^2+y^2+z^2}} & -\frac{cz}{\sqrt{x^2+y^2+z^2}} \end{bmatrix} = -\frac{c\mathbf{r}}{\|\mathbf{r}\|^3}$$

Example 13.10

Show that $\mathbf{F} = (3x^2y^2 - 3x^2)\mathbf{i} + (2x^3y + 2)\mathbf{j}$ is conservative and find its potential.

Find $\varphi: \mathbb{R}^2 \rightarrow \mathbb{R}$ s.t. $\mathbf{F} = \nabla \varphi$

$$F_1 = \frac{\partial \varphi}{\partial x}$$

$$F_2 = \frac{\partial \varphi}{\partial y}$$

$$\frac{\partial}{\partial x} \varphi(x, y) = 3x^2y^2 - 3x^2$$

$$\Rightarrow \varphi(x, y) = x^3y^2 - x^3 + f(y)$$

$$\frac{\partial}{\partial y} \varphi(x, y) = 2x^3y + 2$$

$$\Rightarrow \varphi(x, y) = x^3y^2 + 2y + g(x)$$

$$\varphi(x, y) = x^3y^2 - x^3 + 2y$$

is a potential of $\mathbf{F}$ ($\mathbf{F} = \nabla \varphi$)

$\Rightarrow \mathbf{F}$ is a conservative vector field

Theorem 13.11 (Cross partials coincide for conservative fields)

If $\mathbf{F} : D \rightarrow \mathbb{R}^3$ is a conservative vector field such that its components F_1, F_2, F_3 have continuous partial derivatives, then $\mathbf{F} = \nabla \varphi$

$$\frac{\partial F_1}{\partial y} = \frac{\partial F_2}{\partial x}, \quad \frac{\partial F_2}{\partial z} = \frac{\partial F_3}{\partial y}, \quad \frac{\partial F_3}{\partial x} = \frac{\partial F_1}{\partial z}.$$

$$F_1 = \frac{\partial \varphi}{\partial x} \quad \frac{\partial}{\partial y} \frac{\partial \varphi}{\partial x} = \frac{\partial}{\partial x} \frac{\partial \varphi}{\partial y}$$

$$F_2 = \frac{\partial \varphi}{\partial y}$$

$$F_3 = \frac{\partial \varphi}{\partial z}$$

Definition 13.12 (Irrotational vector field)

A vector field $\mathbf{F} : D \rightarrow \mathbb{R}^3$ satisfying

$$\operatorname{curl} \mathbf{F} = \nabla \times \mathbf{F} = 0$$

is called irrotational.

It follows from Theorem 13.11 that conservative fields are irrotational,

$$\operatorname{curl} \mathbf{F} = \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right) \mathbf{i} + \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \right) \mathbf{j} + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \mathbf{k} = 0.$$

Note that the converse is only true if the function $\mathbf{F}$ is considered on a simply connected domain.

conservative $\tilde{\mathbf{F}} = \nabla \varphi$
 $\Rightarrow$ irrotational $\cdot \operatorname{curl} \tilde{\mathbf{F}} = 0$
not irrotational $\Rightarrow$ not conservative

Example 13.13

Show that $\mathbf{F} = e^x \mathbf{i} + y^2 \mathbf{j} + xz \mathbf{k}$ is not a conservative field.

$$\text{curl } \mathbf{F} = \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ e^x & y^2 & xz \end{vmatrix} = (0 - 0) \mathbf{i} + (0 - z) \mathbf{j} + (0 - 0) \mathbf{k} \\ = -z \mathbf{j} \neq \mathbf{0}$$

$\Rightarrow$ not irrotational $\Rightarrow$ not conservative.

Definition 13.14 (Laplacian)

The Laplacian $\Delta = \nabla \cdot \nabla$ of a twice continuously differentiable function f is

$$\underline{\Delta f} = \underline{\operatorname{div} \nabla f} = \underline{\nabla \cdot \nabla f}.$$

Explicitly, we have for $f(x, y, z)$ and $f(x, y)$:

$$\underline{\Delta f(x, y, z) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2};}$$

$$\underline{\Delta f(x, y) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}.}$$

Example 13.15

Let $f(x, y, z) = e^{xyz}$. Calculate Δf .

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial}{\partial x} e^{xyz} \right) = \frac{\partial}{\partial x} (e^{xyz} \cdot yz) = e^{xyz} \cdot y^2 z^2, \text{ similar to } \frac{\partial^2 f}{\partial y^2}, \frac{\partial^2 f}{\partial z^2}$$
$$\therefore \Delta f = e^{xyz} (y^2 z^2 + x^2 y^2 + x^2 z^2)$$